

## Number, units and estimation - Set 1

TMUA practice questions by topic. Specification M1.1, M1.2, M2.1-M2.14. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the highest common factor of  $2^3 \times 3^2 \times 5$  and  $2^2 \times 3^3 \times 7$ .

A 18  
B 6  
C 12  
D 36  
E 72  
F 108

- 2 Express the recurring decimal  $0.\dot{2}\dot{7}$  as a fraction in its lowest terms.

A  $\frac{9}{33}$   
B  $\frac{18}{33}$   
C  $\frac{27}{11}$   
D  $\frac{99}{100}$   
E  $\frac{107}{100}$   
F  $\frac{27}{100}$

- 3 A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

A  $16.1 \text{ cm}^2$   
B  $11.375 \text{ cm}^2$   
C  $16.45 \text{ cm}^2$   
D  $16.275 \text{ cm}^2$   
E  $14.375 \text{ cm}^2$   
F  $13.8 \text{ cm}^2$

- 4 Evaluate  $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$ , giving your answer in standard form.
- A**  $4 \times 10^4$   
**B**  $4 \times 10^6$   
**C**  $4 \times 10^{-1}$   
**D**  $2.4 \times 10^5$   
**E**  $1.5 \times 10^5$   
**F**  $4 \times 10^5$
- 5 A code consists of two letters from A, B, C, D, E followed by two digits from 1 to 9. Letters may be repeated but the two digits must be different. How many codes are possible?
- A** 3240  
**B** 2025  
**C** 1440  
**D** 900  
**E** 1620  
**F** 1800
- 6 A car travels at 90 km/h. What is this speed in metres per second?
- A** 30 m/s  
**B** 54 m/s  
**C** 15 m/s  
**D** 25 m/s  
**E** 324 m/s  
**F** 1.5 m/s
- 7 Estimate the value of  $\frac{4.92 \times \sqrt{99.3}}{0.198}$  by rounding each number to 1 significant figure.
- A** 2500  
**B** 250  
**C** 500  
**D** 100  
**E** 50  
**F** 25

- 8** A number  $x$  is truncated (not rounded) to 2 decimal places and the result is 3.14. Which inequality describes the possible values of  $x$ ?
- A**  $3.14 < x < 3.15$
- B**  $3.135 \leq x < 3.145$
- C**  $3.13 < x \leq 3.14$
- D**  $3.135 < x < 3.145$
- E**  $3.14 \leq x < 3.15$
- F**  $3.14 \leq x \leq 3.15$
- 9** Evaluate  $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$ .
- A**  $\frac{2}{37}$
- B**  $\frac{13}{72}$
- C**  $\frac{49}{72}$
- D**  $\frac{2}{7}$
- E**  $\frac{24}{24}$
- F** 1
- 10** Which of the following is equal to  $\frac{8^4 \times 2^{-5}}{4^3}$ ?
- A** 1
- B** 2
- C** 16
- D** 8
- E** 4
- F**  $\frac{1}{2}$
- 11** Simplify  $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$ .
- A** 18
- B** 12
- C**  $9\sqrt{3}$
- D** 6
- E** 36
- F**  $6\sqrt{3}$

**12** How many three-digit positive integers have digits that are all different and all odd?

**A** 120

**B** 24

**C** 125

**D** 100

**E** 30

**F** 60